Screen Time vs Mental Health Among Swedish Teenagers

An Exploratory Data Analysis on Sleep Quality Patterns and BDI Depression Indicators

Presented by:
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About Me

A Statistics graduate from the Islamic University of Indonesia (GPA 3.63/4.00) with a strong interest in data analysis and artificial intelligence.

Possesses internship experience at Bappedalitbang Majalengka in data preparation, analysis, visualization, dashboard development and research for Majalengka macroeconomic. Also developed a Retrieval-Augmented Generation (RAG)-based chatbot for the Statistics Department website using N8N tool to integrate Large Language Models (LLMs) with a vector database for intelligent information retrieval.

Proficient in RStudio, Microsoft Excel, Power BI, Tableau, Python and SQL.

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Introduction

Background

The use of digital devices (screen time) among adolescents has risen sharply, sparking concerns about its impact on mental health and sleep patterns.

Problem

Is increased screen time duration directly associated with reduced sleep quality and an increased risk of depression among Swedish adolescents?

Dataset

Using survey data from 4,810 Swedish adolescents, sourced from Kaggle, covering demographic variables, screen time index, sleep duration, and Beck Depression Inventory (BDI) scores to explore the relationship between screen time, sleep, and mental health.

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Project Overview

Purpose

1. Map the distribution of screen time by gender.
2. Measure the effect of screen time on average sleep duration and BDI scores.
3. Analyze the proportion of depression levels within each screen time usage group.

Methodology

1. Excel Power Query: cleaning decimal data and standardizing categories.
2. Pivot Table with data aggregation and percentage calculations.
3. Interactive dashboard featuring interactive visualization using slicers.

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Key Performance Indicator

1 Total Respondent

the number of Swedish adolescent respondents surveyed.

2 Average Screen Time Index

average screen time duration across all respondents.

3 Average Sleep Hours

respondents' average daily sleep duration in hours.

4 Average Depression Score

average depression level score of respondents using the Beck Depression Inventory (BDI) scale.

5 Depression Rate

the proportion of total respondents identified as depressed.

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Discussion Section

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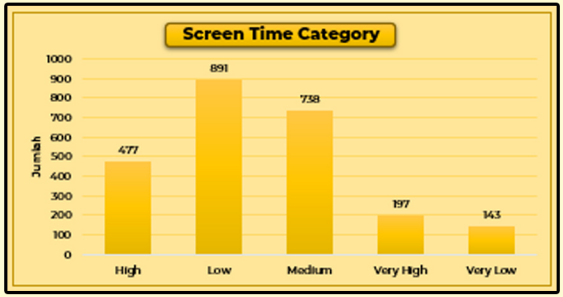
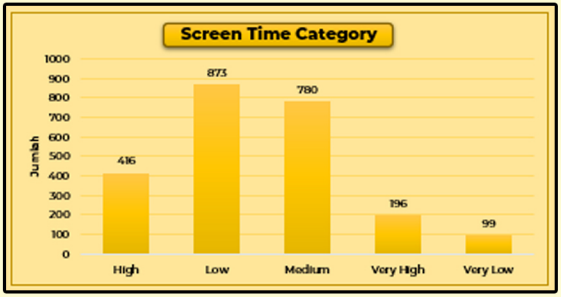
Screen Time by Gender





Data visualization regarding screen time categories among male and female respondents reveals a relatively similar distribution pattern, with the "Low" and "Medium" categories predominating for both genders. Specifically, 891 of males and 873 of females fall into the "Low" category, while the "Medium" category comprises 738 of males and 780 of females. A notable difference appears in the "High" usage category, where the male group shows a higher count 477 respondents compared to the female group, which has 416 respondents. Meanwhile, in extreme categories such as "Very High," the proportions of the two groups tend to be balanced, with a difference of one additional male. However, in the "Very Low" category, the number of male respondents is slightly higher 143 than females 99.

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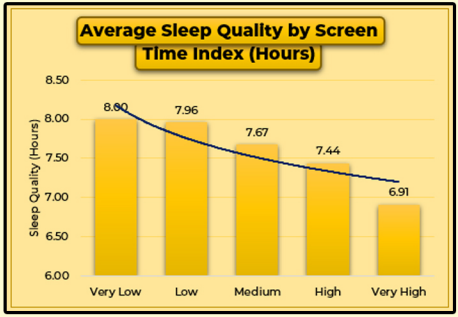
Screen Time by Average Sleep Quality



Based on the visualization above, there is a clear downward trend linking respondents' daily sleep duration to their screen time. The most drastic decline is observed in the "Very High" group, where the average sleep duration decreases by 0.53 hours compared to the "High" group. Overall, the graph indicates that excessive use of digital devices can significantly disrupt daily sleep duration.

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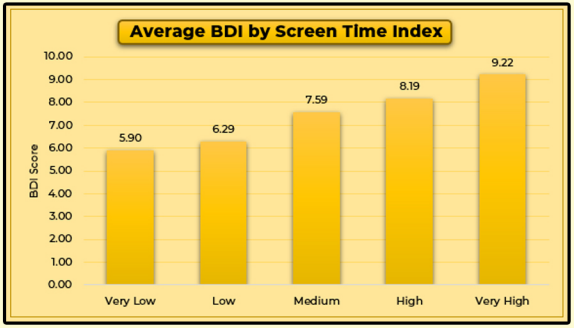
Screen Time by Average BDI Score



Based on the graph above, a direct relationship is evident between screen time duration and the respondents' average BDI scores. Adolescents in the "Very Low" and "Low" categories recorded the lowest average BDI scores 5.90 and 6.29, respectively both of which remain below the 13 point threshold indicating depression. As screen time increased, so did the average BDI scores across the "Medium" (7.59), "High" (8.19), and "Very High" (9.22) categories, although all groups remained below the 13-point cutoff. Overall, the 3.32 point increase indicates that higher screen time correlates with higher average BDI scores and a greater likelihood of depressive symptoms.

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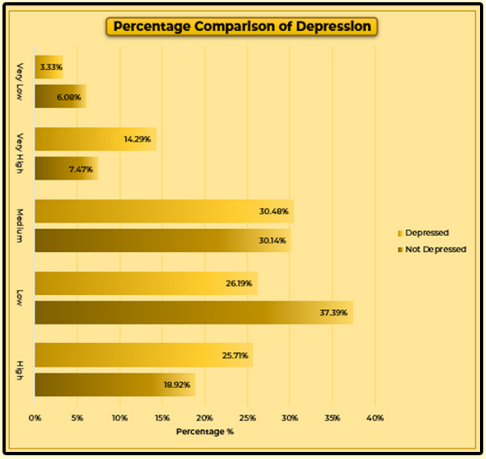
Percentage Comparison of Depression



The graph illustrates a difference in screen usage patterns between respondents experiencing depression and those who are not. In the depression group, screen usage tends to be higher, with the "High" and "Very High" categories accounting for 40.00%, compared to 26.39% in the non-depressed group. Conversely, a larger proportion of non-depressed respondents fall into the "Low" and "Very Low" categories, totaling 43.47%. Meanwhile, the "Medium" category shows a relatively similar proportion across both groups, at approximately 30%. Overall, these results indicate that higher screen usage tends to be associated with a greater prevalence of depression among the respondents.

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Conclusion

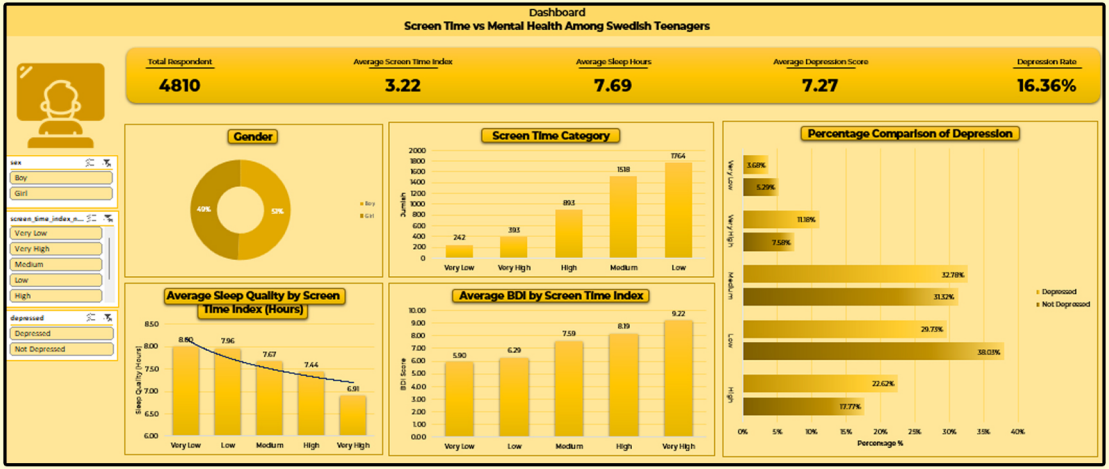
- Male adolescents predominate in several screen-time categories compared to females. A significant difference is observed only in the medium category, where the number of females exceeds that of males by 42.

- There is a downward trend in daily sleep duration as screen usage increases, dropping from 8.00 hours (Very Low) to 6.91 hours (Very High). The sharpest decline a drop of 0.53 hours occurs between the High and Very High categories.

- The average Beck Depression Inventory (BDI) score consistently increased by 3.32 points, rising from 5.90 (Very Low) to 9.22 (Very High). Although all averages remained below the depression threshold (13 points), higher levels of screen time correlated with an increase in depressive symptoms.

- Among adolescents showing signs of depression, 40.00% fell into the high screen usage categories (25.71% "High" and 14.29% "Very High"). This figure is significantly higher than that of the non-depressed group, where the combined percentage for these two categories was only 26.39%.

Dashboard


Dashboard
Screen Time vs Mental Health Among Swedish Teenagers
Total Respondent
4810
Average Screen Time Index
3.22
Average Sleep Hours
7.69
Average Depression Score
7.27
Depression Rate
16.36%
sex
Boy
Girl
screen_time_index_n...
Very Low
Very High
Medium
Low
High
depressed
Depressed
Not Depressed
Gender
49%
51%
Boy
Girl
Screen Time Category
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
242
393
893
1538
1764
Very Low
Very High
High
Medium
Low
Percentage Comparison of Depression
Very Low
Very High
Medium
Low
High
3.69%
5.29%
11.18%
7.58%
32.78%
31.32%
29.75%
38.03%
22.62%
17.77%
Depressed
Not Depressed
0%
5%
10%
15%
20%
25%
30%
35%
40%
Percentage %
Average Sleep Quality by Screen Time Index (Hours)
Sleep Quality (Hours)
8.50
8.00
7.50
7.00
6.50
6.00
8.00
7.96
7.67
7.44
6.91
Very Low
Low
Medium
High
Very High
Average BDI by Screen Time Index
BDI Score
10.00
9.00
8.00
7.00
6.00
5.00
4.00
3.00
2.00
1.00
0.00
5.90
6.29
7.59
8.19
9.22
Very Low
Low
Medium
High
Very High

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